

# The angle in a semicircle

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Circle Theorems · Lesson 3. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

## What this lesson is about

This is the first and cleanest consequence of last lesson's keystone. Let the two points A and B be the ends of a **diameter** — a chord straight through the centre O. Then the arc AB is a **semicircle**, and the angle at the centre  $\angle AOB$  is a **straight line =  $180^\circ$** . Since the angle at the circumference is half the angle at the centre,  $\angle APB = \frac{1}{2} \times 180^\circ = 90^\circ$ . So **every triangle drawn on a diameter has a right angle** at its third vertex — and it stays  $90^\circ$  wherever P sits on the semicircle. This is often called **Thales' theorem**. It also works in reverse (the **converse**): if an angle at the circumference is  $90^\circ$ , the chord it stands on must be a diameter.

## What you are learning to notice

- The **angle in a semicircle is  $90^\circ$**  — an angle drawn on a diameter is a right angle.
- It is the special case of the keystone: a diameter gives  $180^\circ$  at the centre, halved to  $90^\circ$ .
- The other two angles of the triangle then add to  **$90^\circ$**  (angles in a triangle sum to  $180^\circ$ ).
- **Converse**: a right angle at the circumference must stand on a **diameter**.

## Moving through it, one step at a time

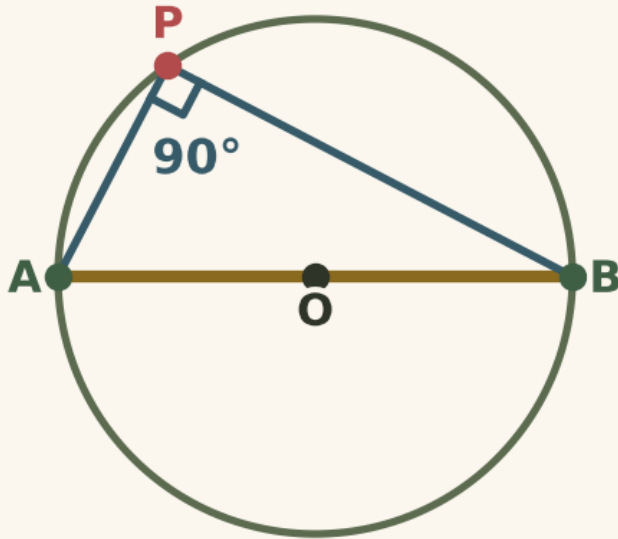
Spot the diameter (a chord through the centre). The angle at the circumference standing on it is  $90^\circ$  — write that with the reason 'angle in a semicircle'. To find another angle in the triangle, use that the three angles sum to  $180^\circ$ , so the two acute angles add to  $90^\circ$ . Always give a reason at each step.

## Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

### Example 1 — Find the missing angle

AB is a diameter and P is on the circle. One acute angle,  $\angle PAB$ , is  $35^\circ$ . Find  $\angle PBA$ .



1. AB is a diameter, so  $\angle APB = 90^\circ$ .      angle in a semicircle

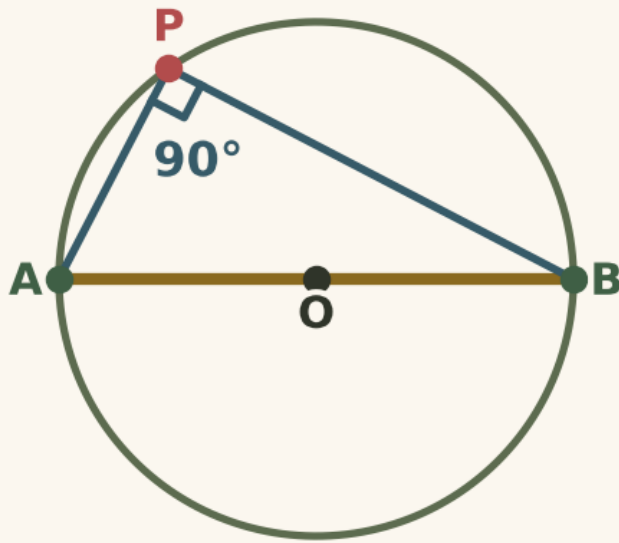
2. Angles in triangle APB sum to  $180^\circ$ .       $35 + 90 + \angle PBA = 180$

3. So the two acute angles add to  $90^\circ$ .       $\angle PBA = 90 - 35$

**$\angle PBA = 55^\circ$ , because the angle in a semicircle is  $90^\circ$  and the angles of the triangle sum to  $180^\circ$ .**

### Example 2 — Why it is a right angle

Prove that the angle in a semicircle is  $90^\circ$ .



1. AB is a diameter, so it passes through the centre diameter through O.
2. A diameter is a straight line, so  $\angle AOB = 180^\circ$ . straight line at O
3. Angle at the centre = 2  $\times$  angle at the circumference.  $\angle APB = \frac{1}{2} \times 180^\circ$

$\angle APB = 90^\circ$ . The angle in a semicircle is a right angle — the special case of the angle-at-the-centre theorem when the arc is a semicircle. ■

### A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

### If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the angle at the centre theorem (Lesson 2), that a diameter passes through the centre, and that angles on a straight line and in a triangle each sum to  $180^\circ$ ), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.

- **Worked solution** — a full model to compare your thinking against.

### How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

## Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

## Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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